Pre-Board Examination (2024-25)
Class – X
Basic Mathematics (241)

Time Allowed: 3 Hrs Maximum Marks: 80

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D, and E.
2. Section A has 20 Multiple Choice Questions (MCQs) carrying 1 mark each.
3. Section B has 5 Short Answer-I (SA-I) type questions carrying 2 marks each.
4. Section C has 6 Short Answer-II (SA-II) type questions carrying 3 marks each.
5. Section D has 4 Long Answer (LA) type questions carrying 5 marks each.
6. Section E has 3 sourced based/Case Based/passage based/integrated units of assessment (4 marks each) with sub-parts of the values of 1, 1 and 2 marks each respectively.
7. All Questions are compulsory. However, an internal choice in 2 Qs of 2 marks, 2 Qs of 3 marks and 2 Questions of 5 marks has been provided. An internal choice has been provided in the 2 marks questions of Section E.
8. Draw neat figures wherever required. Take π =22/7 wherever required if not stated.

SECTION A

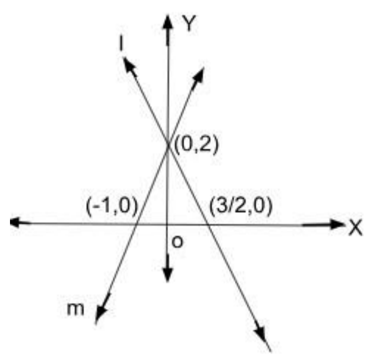
1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; where x, y are prime numbers, then HCF (a,b) is:

a) xy b) xy^2 c) x^3y^3 d) x^2y^2

2. The LCM of smallest two digit composite number and smallest composite number is:

a) 12 b) 4 c) 20 d) 44

3 The system of linear equations represented by the lines l and m is



(A) consistent with unique solution (B) inconsistent
(C) consistent with three solutions (D) consistent with many solutions

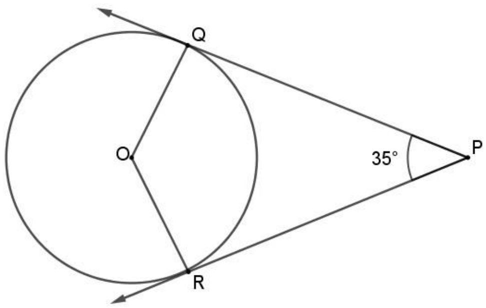
4 Value(s) of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is :

a. 0 only b) 4 c) 8 only d) 0,8

5 The distance of the point(3, 5) from x-axis is k units, then k equals:

a. 3 b) 4 c) 5 d) 8

6 In the given figure, PQ and PR are tangents to a circle centred at O. If $\angle QPR = 35^\circ$ then $\angle QOR$ is equal to



(A) 70° (B) 90° (C) 135° (D) 145°

7. If $\triangle ABC \sim \triangle PQR$ such that 3AB = 2PQ and BC=10 cm, then length QR is equal to

(A) 10 cm (B) 15 cm (C) $\frac{20}{3}$ cm (D) 30 cm

8. If the height of the tower is equal to the length of its shadow, then the angle of elevation of the sun is ______

(a) 30° b) 45° c) 60° d) 90°

9. $(1 - \cos^2 A)$ is equal to

(b) $\sin^2 A$ b) $\tan^2 A$ c) $1 - \sin^2 A$ d) $\sec^2 A$

10 The radius of a circle is same as the side of a square. Their perimeters are in the ratio

(c) 1 : 1 b) $2 : \pi$ c) $\pi : 2$ d) $\sqrt{\pi} : 2$

11 Value of "p" from the polynomial $x^2 + 3x + p$, if one of the zeroes of the polynomial is 2.

a. -10 b) 14 c) 10 d) 17

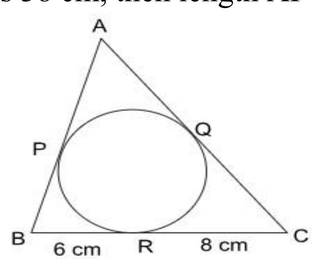
12. When a dice is thrown once, the probability of getting an even number less than 4 is

b. 1/4 b) 0 c) 1/2 d) 1/6

13, The 4^{th} term from the end of the A.P $-8, -5, -2, \dots, 49$ is

(A) 37 (B) 40 (C) 1 (D) 43

14, If perimeter of given triangle is 38 cm, then length AP is equal to



(A)19 cm (B) 5 cm (C) 10 cm (D)8 cm

15. Zeroes of the polynomial $y^2 - 100 = 0$ are,

(A) 4, -4 (B) 100, -100 (C) 10, -10 (D) 10, 20

16. The roots of quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$ are

(A) not real (B) real and equal

(C) rational and distinct (D) irrational and distinct

17.For the following distribution:

Class	0-5	5-10	10-15	15-20	20-25
Frequency	10	15	12	20	9

The lower limit of modal class is:

a. 15 b) 20 c) 10 d) 5

18.The following distribution shows the marks distribution of 80 students.

Marks	Below 10	Below 20	Below 30	Below 40	Below 50	Below 60
No. of students	2	12	28	56	76	80

The median class is

(A) 20-30 (B) 40-50 (C) 30-40 (D) 10-20

DIRECTIONS FOR QUESTION 19 AND 20

a). Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

b). Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

c). Assertions (A) is true but reason (R) is false.

d). Assertions (A) is false but reason (R) is true.

19. Assertion(A): $(2+\sqrt{3})\sqrt{3}$ is an irrational number.

Reason(R): Product of two irrational numbers is always irrational.

20.Assertion (A): The point (0, 4) lies on y-axis.

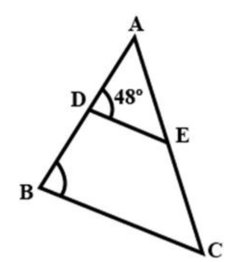
Reason(R): The x-coordinate of a point on y-axis is zero

SECTION B

21.Find whether the following pair of linear equations is consistent or inconsistent:

$$3x + 2y = 8$$

$$6x - 4y = 9$$



22. In figure, if AD = 6cm, DB = 9cm, AE = 8cm and EC = 12cm and $\angle ADE = 48^\circ$. Find $\angle ABC$.

23. The length of a tangent from a point A at distance 5cm from the centre of the circle is 4cm. Find the radius of the circle.

24. Evaluate: $\sin^2 60^\circ + 2\tan 45^\circ - \cos^2 30^\circ$.

25. Find the diameter of a circle whose area is equal to the sum of the areas of two circles of radii 40cm and 9cm.

OR

A chord of a circle of radius 10cm subtends a right angle at the centre. Find the area of minor segment. (Use $\pi = 3.14$)

Section C

(Section C consists of 6 questions of 3 marks each.)

26. Prove that $\sqrt{5}$ is an irrational number.

27. Find the sum of the first 22 terms of an AP in which $d = 7$ and 22nd term is 149.

OR

Which term of the AP: 21, 18, 15, . . . is – 81?

28. Find the mean using the step deviation method.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	6	10	15	9	10

29. Prove that sec A (1 – sin A) (sec A + tan A) = 1.

30. The sum of a two-digit number and the number obtained by reversing the order of its digits is 99. If ten's digit is 3 more than the unit's digit, then find the number.

31. A circle touches all the four sides of quadrilateral ABCD. Prove that AB + CD = AD + BC.

Section D consists of 4 questions of 5 marks each

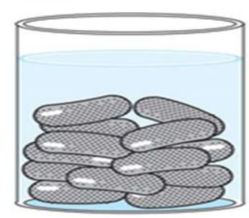
32. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

33.From the top of a 9 m high building, the angle of elevation of the top of a cable tower is 60° and angle of depression of its foot is 45°. Determine the height of the tower and distance between building and tower. (Use $\sqrt{3} = 1.732$)

34.A medicine capsule is in the shape of a cylinder with two hemispheres stuck at each of its ends. The length of the entire capsule is 14mm and the diameter of the capsule is 5mm. Find its surface area.

OR

A gulab jamun, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like cylinder with two hemispherical ends with length 5cm and diameter 2.8cm.



35. The following table gives the distribution of the life time of 400 neon lamps:

Life time (in hours)	Number of lamps
1500-2000	14
2000-2500	56
2500-3000	60
3000-3500	86

3500-4000	74
4000-4500	62
4500-5000	48

Find the average life time of a lamp.

SECTION E (4 MARKS EACH)

36. Raj and Ajay are very close friends. Both the families decide to go to Shimla by their own cars. Raj's car travels at a speed of x km/h while Ajay's car travels 5 km/h faster than Raj's car. Raj took 4 hours more than Ajay to complete the journey of 400 km.

(i) . What will be the distance covered by Ajay's car in two hours?
(ii). Write quadratic equation to describe the speed of Raj's car?
(iii). What is the speed of Raj's car?

Or

How much time took Ajay to travel 400 km?

37. Rinku was very happy to receive a fancy jumbo pencil from his best friend Rohan on his birthday. Pencil is a basic writing tool, when sharpened its shape is a combination of cylinder & cone as given in the picture.

Cylindrical pencil with conical head is a common shape worldwide since ages. Commonly pencils are made up of wood & plastic but we should promote pencils made up of eco-friendly material (many options available in the market these days) to save environment.

The dimensions of Rinku's pencil are given as follows:

Length of cylindrical portion is 21cm. Diameter of the base is 1 cm and height of the conical portion is 1.2 cm

Based on the above information, answer the following questions

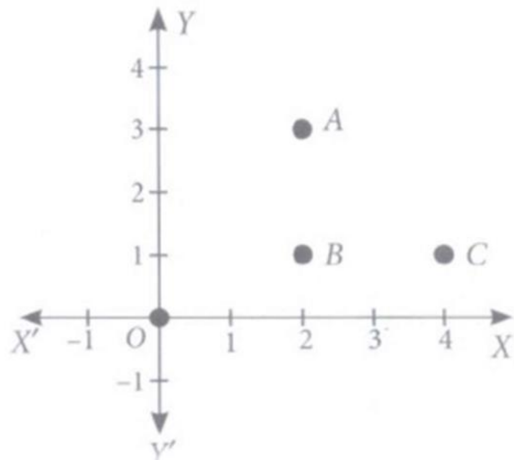
1. Find the slant height of the sharpened part. (1)

2. Find curved surface area of sharpened part (in terms of π). (1)
3. Find the total surface area of the pencil (in terms of π). (2)

OR

The pencil's total height decreases by 8.2 cm after sharpening it many times, what is the volume of the cylindrical part of the shortened pencil (in terms of π)?

38. Alia and Shagun are friends living on the same street in Patel Nagar. Shagun's house is at the intersection of one street with another street on which there is a library. They both study in the same school and that is not far from Shagun's house. Suppose the school is situated at the point O, i.e., the origin, Alia's house is at A. Shagun's house is at B and library is at C.
Based on the above information, answer the following questions.



(i) How far is Alia's house from Shagun's house?
(ii) How far is the library from Shagun's house?
(iii) Show that for Shagun, school is farther compared to Alia's house and library.

OR

Show that Alia's house, shagun's house and library for an isosceles right triangle.